Minimax Optimal (Parks-McClellan) FIR Filter Design Examples

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Key Concepts

1) MATLAB provides a function that designs minimax optimal linear phase FIR filters using the Parks-McClellan algorithm.

    a) Frequency bands must be specified.

        i) Normalized frequency is used where $[0, 1]$ corresponds to ω on $[0, \pi]$ rads.

        ii) Each band is defined by a start frequency and stop frequency.

        iii) The number of frequencies must be even.

b) Gains in each band must be specified.

    i) The gain at the start and stop frequency for each band must be specified.

    ii) The number of gains must be even.

    iii) The response in a band is the line connecting the gain at the start frequency to the gain at the stop frequency.

c) The relative weight on the error in each band can be specified.

    i) The number of weights is equal to the number of bands - one-half the number of frequencies.

ii) The weights specify the relative importance of the error in each band.

iii) The relative size of the ripples is equal to the relative size of the weights.

d) Filter order must be specified.

2) Guess-and-check is used to find a filter order that satisfies the constraints.

Video Screenshots

0:08

Minimax design of FIR filters

- linear phase

- design criterion

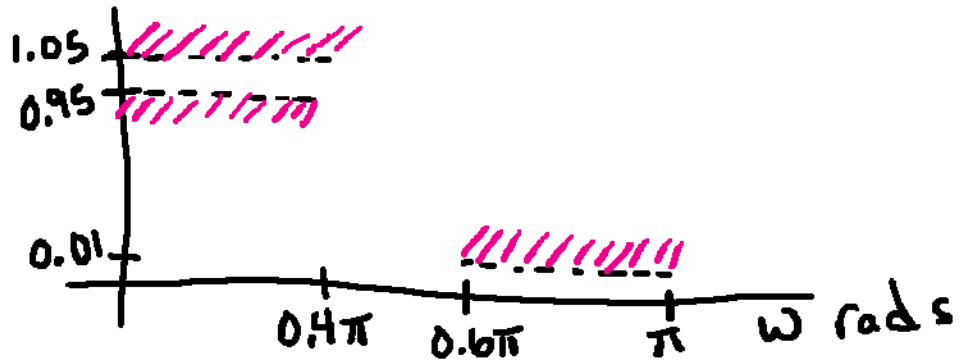
  impulse response $h[n]$, $0 \le n \le M$

$$\min_{h[n]} \left\{ \max_{\omega \in F} \left| W(\omega) \left(H_d(\omega) - A(\omega) \right) \right| \right\}$$

- equiripple weighted error

Low Pass FIR Filter

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Use MATLAB firpm

```
del_p = 0.05;
del_s = 0.01;
om_p = 0.4*pi;
om_s = 0.6*pi;

M = ceil((-10*log10(del_p*del_s) - 13)/(2.324*(om_s-om_p)))

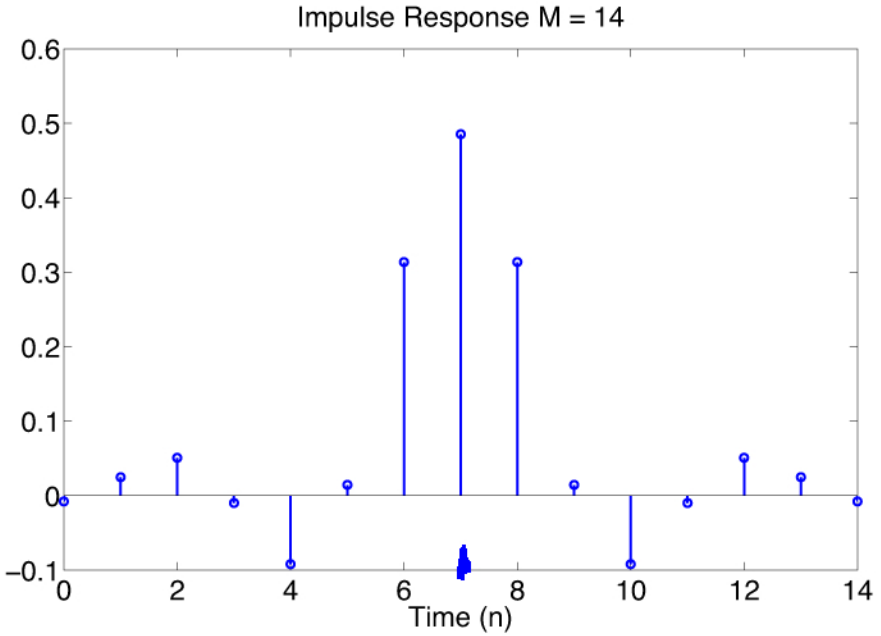
F = [0  om_p  om_s  pi]/pi;
A = [1  1  0  0];
W = [ 1  del_p/del_s];

b = firpm(M, F, A, W);
```

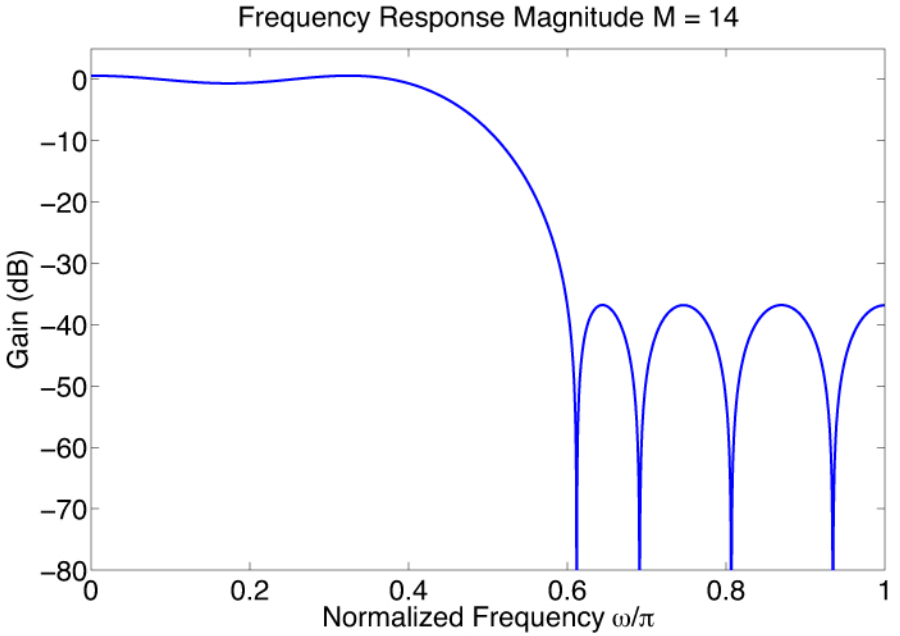
$M = 14$

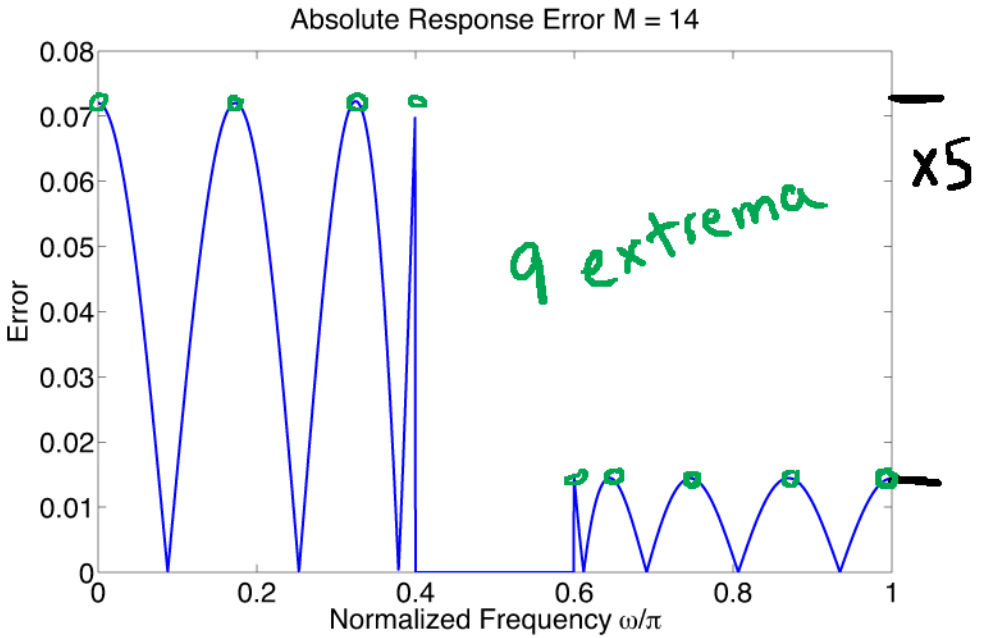
$\frac{0.05}{0.01} = 5 \Rightarrow$ stopband error 5 x as important as passband error

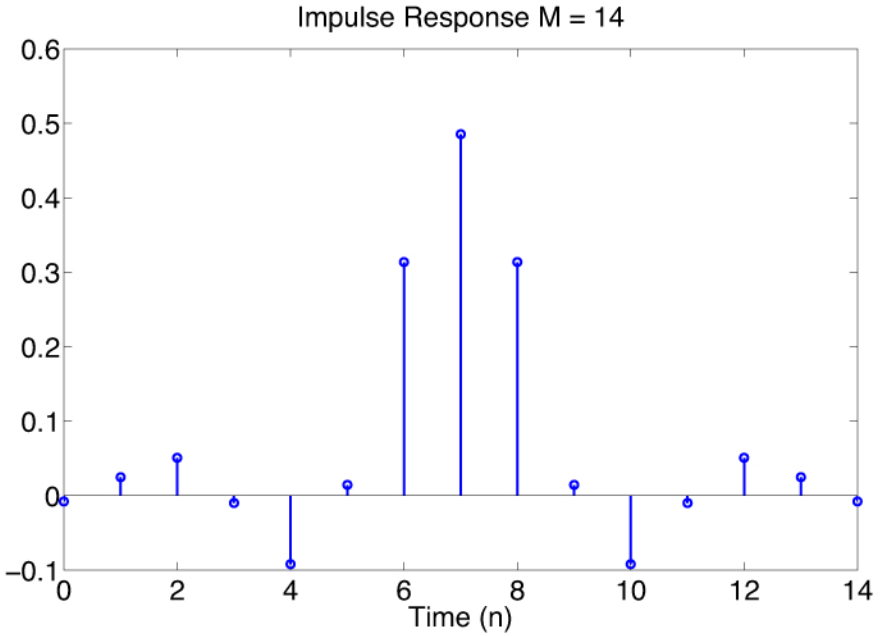
5:14

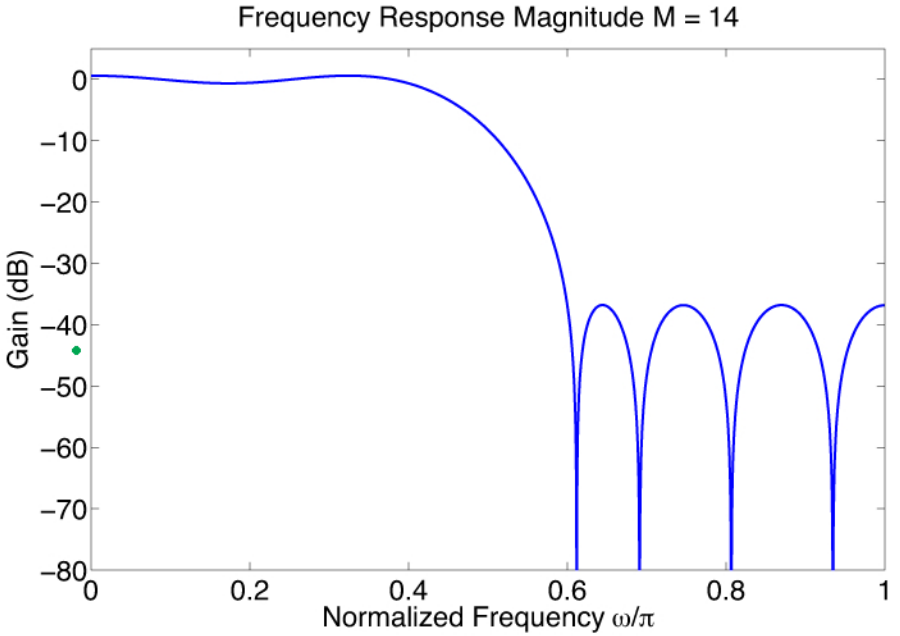

Impulse Response M = 14
0.6
0.5
0.4
0.3
0.2
0.1
0
−0.1
0
2
4
6
8
10
12
14
Time (n)

2


Frequency Response Magnitude M = 14
0
−10
−20
−30
−40
−50
−60
−70
−80
Gain (dB)
0
0.2
0.4
0.6
0.8
1
Normalized Frequency ω/π


Absolute Response Error M = 14
0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0
Error
0
0.2
0.4
0.6
0.8
1
Normalized Frequency ω/π
9 extrema
x5



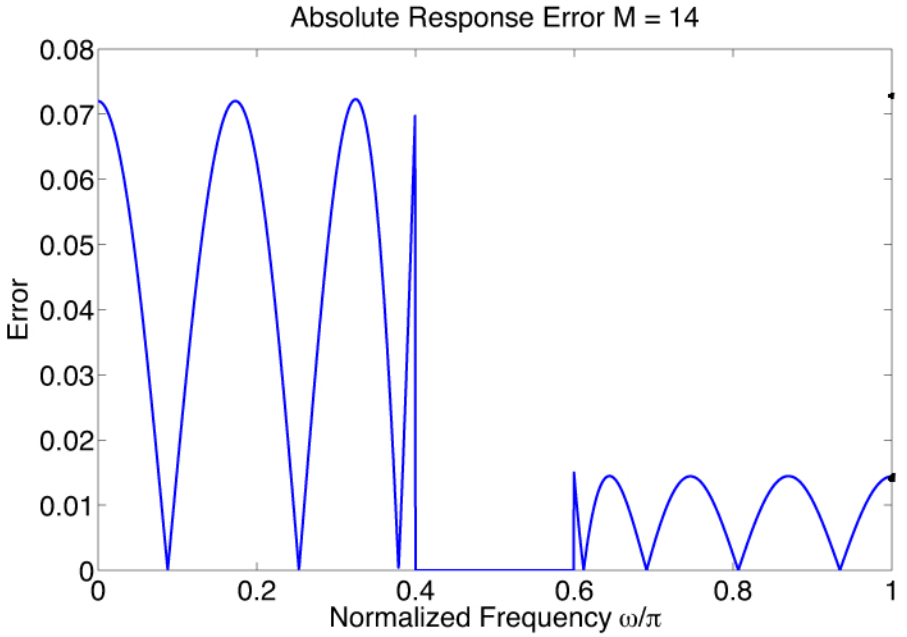


Passband error > 0.05

Stopband error > 0.01

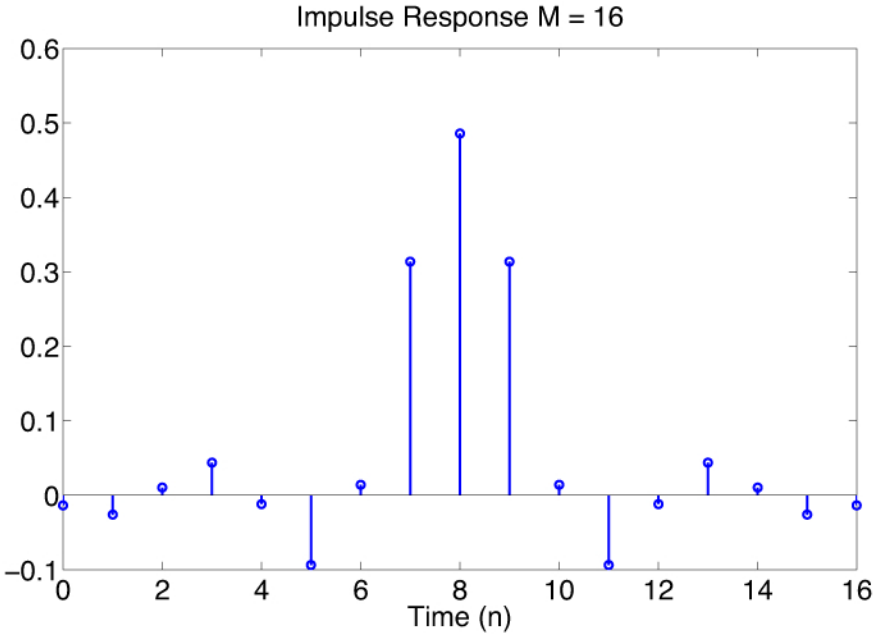
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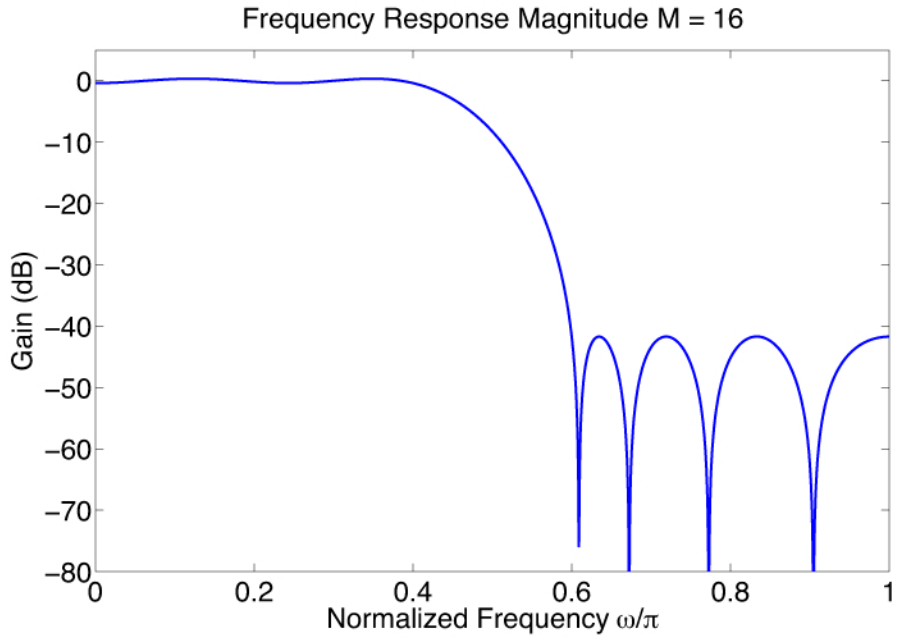
increase M



Try M = 16

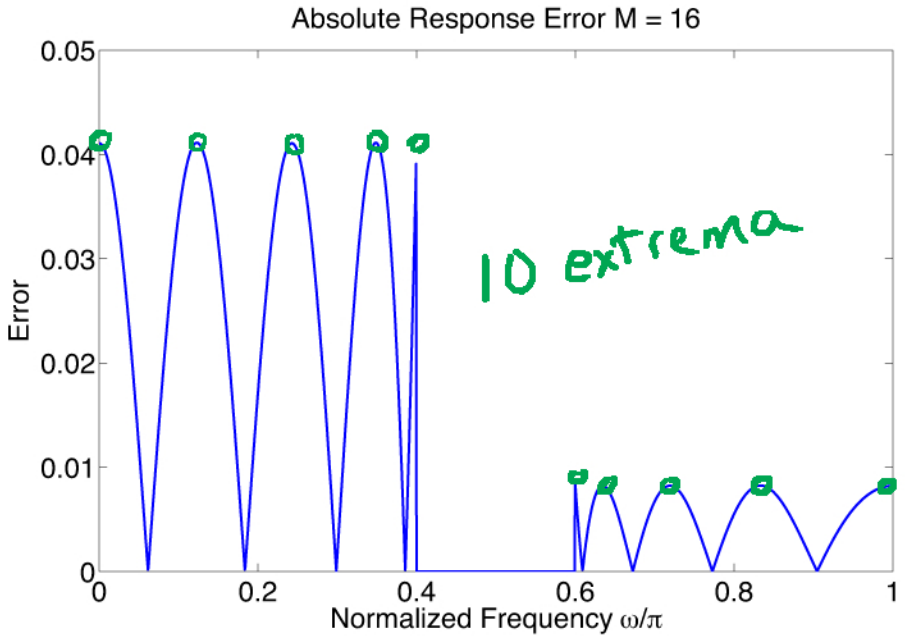
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Passband ripple < 0.05

Stopband ripple < 0.01



Band pass FIR Filter - different weightings

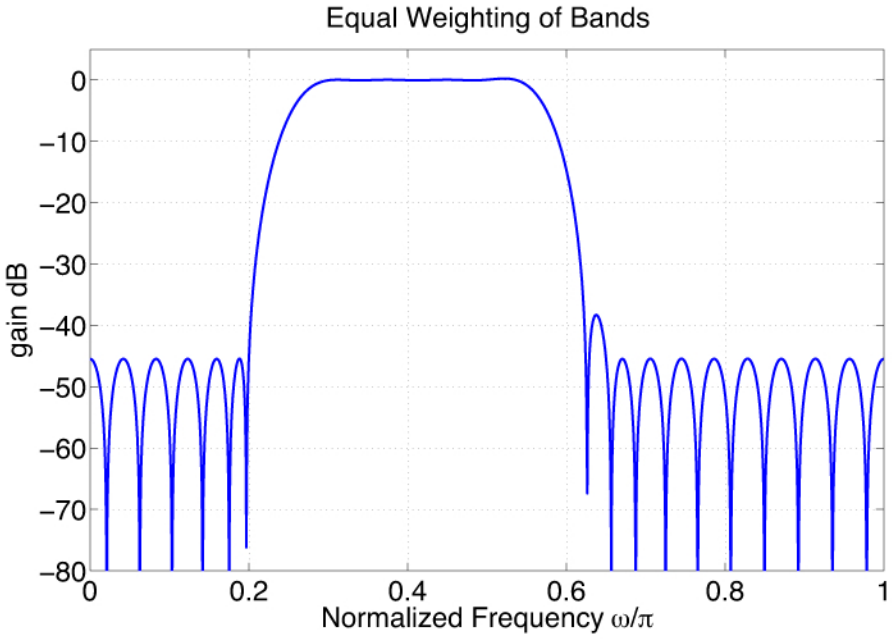
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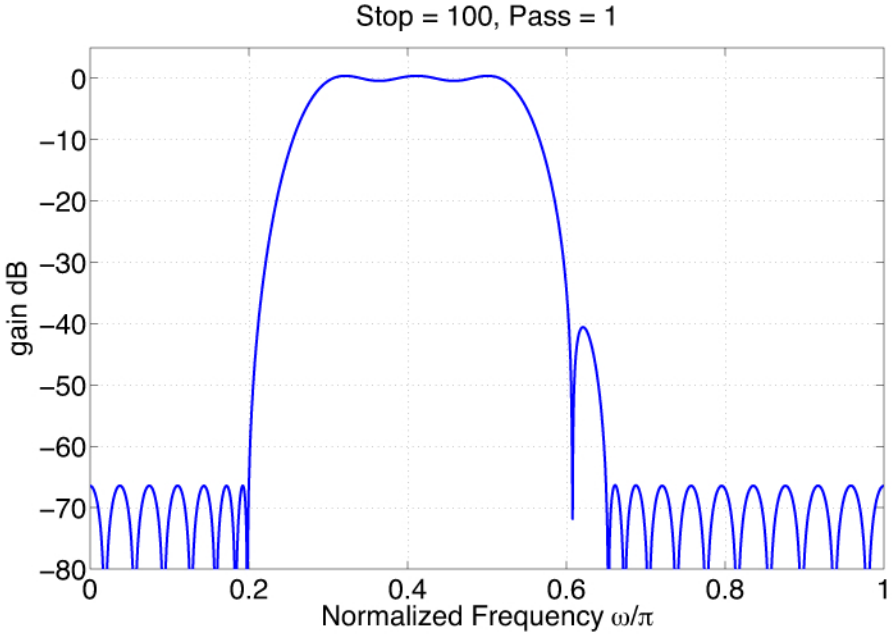
```
order = 46;

freqbands = [0  0.2*pi  0.3*pi  0.5*pi  0.65*pi  pi]/pi;
bandgains = [ 0  0  1  1  0  0];

bandwghts1 = [ 1  1  1];
Bf1 = firpm(order,freqbands,bandgains,bandwghts1);

bandwghts2 = [ 100 1 100];
Bf2 = firpm(order,freqbands,bandgains,bandwghts2);
```

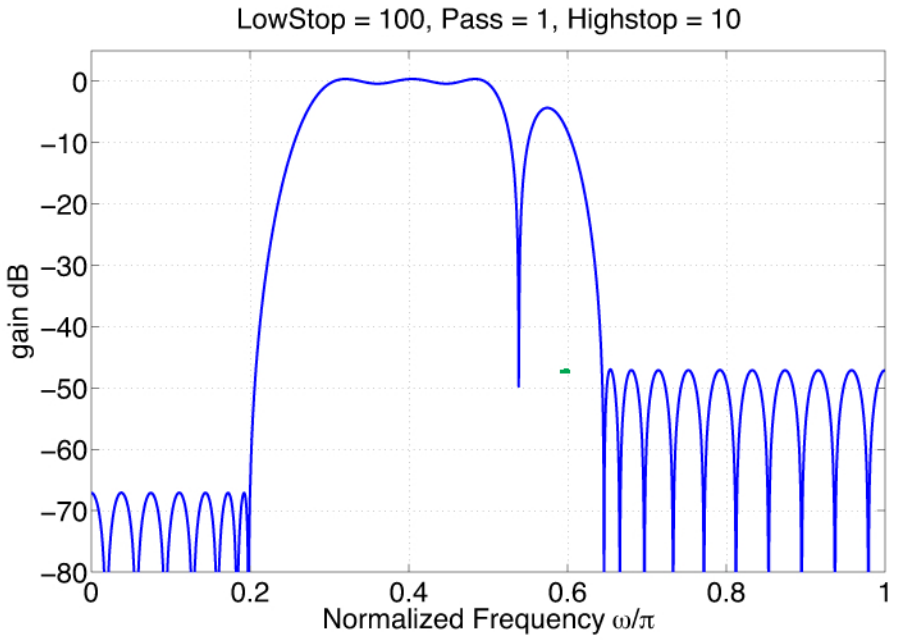


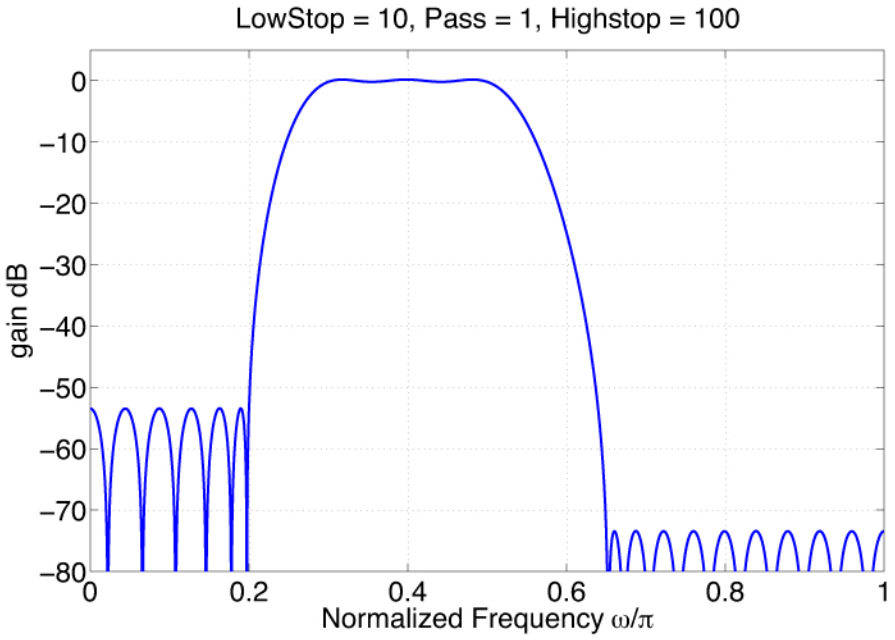


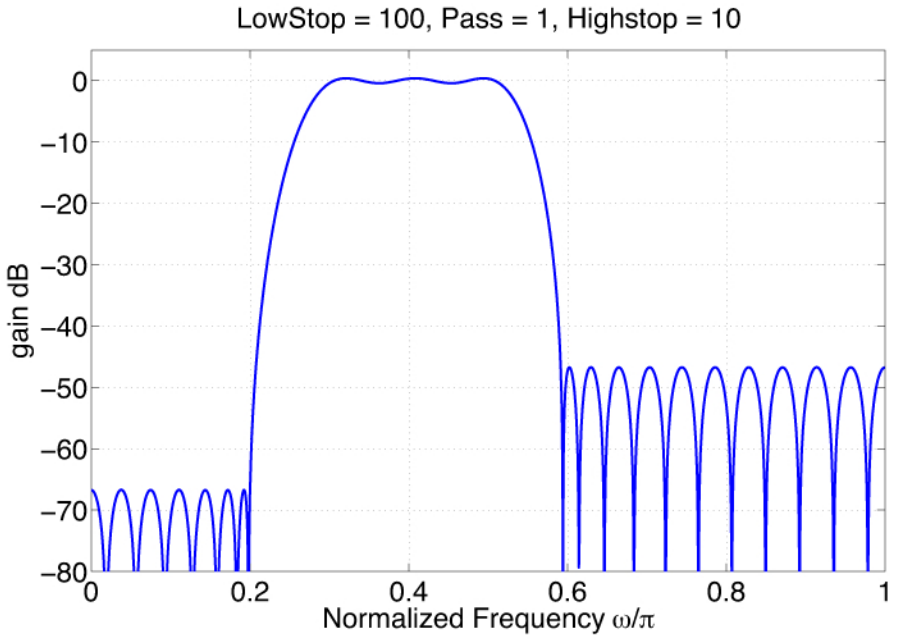
```
bandwghts3 = [10   1   100];
Bf3 = firpm(order,freqbands,bandgains,bandwghts3);

bandwghts4 = [100   1   10];
Bf4 = firpm(order,freqbands,bandgains,bandwghts4);


freqbands = [0   0.2*pi   0.3*pi   0.5*pi   0.6*pi   pi]/pi;
Bf5 = firpm(order,freqbands,bandgains,bandwghts4);
```







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